

MEACHAM, OR, USA

WMO: 726885

Lat: 45.511N

Lon: 118.425W

Elev: 1135

StdP: 88.41

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24152

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-18.8	-14.6	-22.0	0.6	-18.5	-17.4	0.9	-13.8	4.8	3.3	4.2	2.2	0.1	230	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	31.0	15.5	29.2	15.0	27.5	14.5	17.0	27.3	16.1	26.6	15.3	25.5	1.8	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.4	11.0	19.2	12.1	10.1	18.2	11.0	9.3	18.0	51.8	27.1	48.8	26.6	46.1	25.5	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.0	3.1	2.8	-26.1	34.1	4.5	1.9	-29.4	35.4	-32.0	36.5	-34.6	37.6	-37.9	38.9		
(5)			-26.4	19.2	4.5	1.7	-29.6	20.5	-32.3	21.5	-34.8	22.4	-38.1	23.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-1.6	-1.1	1.6	4.6	8.6	11.8	15.8	15.0	11.1	5.9	1.5	-2.6		
	DBStd	7.47	4.84	5.03	3.89	3.42	3.67	3.54	3.04	3.24	3.74	3.92	5.01	5.10		
	HDD10.0	2003	359	311	262	164	72	19	1	2	29	136	258	391		
	HDD18.3	4548	617	544	520	411	302	198	88	112	217	384	506	649		
	CDD10.0	518	0	0	1	3	28	73	182	156	63	9	2	0		
	CDD18.3	22	0	0	0	0	0	2	11	8	1	0	0	0		
	CDH23.3	1729	0	0	0	1	38	136	718	624	207	6	0	0		
	CDH26.7	535	0	0	0	0	6	34	239	202	53	0	0	0		
(14) Wind	WSAvg	0.8	0.7	0.7	0.8	0.9	0.9	0.9	0.8	0.8	0.7	0.6	0.7	0.7		
(15) Precipitation	PrecAvg	693	87	62	73	68	65	49	14	16	24	54	92	88		
	PrecMax	874	122	108	103	110	146	97	60	54	71	88	175	161		
	PrecMin	550	45	30	30	39	15	12	3	2	6	26	28	51		
	PrecStd	89	22	20	21	20	30	18	13	15	16	18	33	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	12.6	16.7	22.2	27.1	30.5	33.3	33.1	30.9	23.9	16.4	9.6		
		MCWB	5.0	6.3	8.3	11.9	14.3	16.4	16.7	16.1	14.1	12.5	8.8	5.8		
	2%	DB	7.1	9.3	13.4	18.7	23.8	27.1	31.1	30.9	28.1	20.1	13.1	6.8		
		MCWB	3.4	4.8	6.5	10.5	13.2	15.3	15.8	15.2	13.7	10.5	7.8	4.0		
	5%	DB	5.4	7.2	10.8	15.6	21.2	24.4	29.4	28.9	25.8	17.3	10.8	5.1		
		MCWB	2.6	3.7	5.4	8.5	12.2	14.0	15.3	14.7	13.0	9.4	6.9	2.9		
	10%	DB	4.0	5.4	8.6	12.6	18.2	22.0	27.6	27.1	23.0	14.4	8.6	3.6		
		MCWB	1.6	2.5	4.3	6.9	10.7	12.8	14.8	14.3	12.2	8.2	5.5	1.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	7.2	9.2	12.9	15.6	18.2	19.2	17.9	16.4	13.4	10.6	6.2		
		MCDB	7.8	10.7	15.0	19.9	24.4	27.9	28.4	29.3	25.7	21.1	14.1	8.6		
	2%	WB	4.2	5.5	7.3	10.8	14.1	16.0	17.4	16.5	15.0	11.5	8.7	4.8		
		MCDB	6.3	8.6	11.7	17.7	22.0	25.1	27.5	27.9	25.0	17.8	12.1	6.4		
	5%	WB	2.9	4.2	5.9	9.0	12.7	14.6	16.3	15.7	13.7	10.2	7.2	3.2		
		MCDB	4.7	6.6	10.1	14.7	19.5	23.1	26.9	26.4	23.5	16.1	10.3	4.6		
	10%	WB	1.9	2.8	4.8	7.4	11.4	13.3	15.4	14.8	12.6	8.9	5.7	1.9		
		MCDB	3.6	4.8	8.3	12.1	17.2	20.7	26.0	25.3	22.0	13.6	8.4	3.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	9.7	11.1	12.3	14.0	15.4	20.2	21.0	19.2	14.2	9.1	7.9		
		MCDBR	9.0	11.6	15.4	19.1	20.8	21.0	23.7	24.7	25.2	20.1	12.2	8.1		
	5% WB	MCWBR	6.3	7.8	9.8	11.6	11.2	10.2	10.1	10.8	12.2	11.6	7.8	5.9		
		MCWBR	8.0	10.0	13.7	18.0	18.0	19.0	21.0	21.7	21.8	17.3	11.1	7.2		
(40) Clear-Sky Solar Irradiance	taub		0.271	0.272	0.284	0.322	0.330	0.325	0.327	0.329	0.315	0.293	0.274	0.265		
		taud	2.520	2.564	2.567	2.439	2.445	2.484	2.491	2.472	2.497	2.562	2.564	2.527		
	Ebn at Noon		859	930	960	939	938	940	934	921	909	888	850	829		
		Edh at Noon	64	74	85	106	109	106	104	102	91	74	61	57		
(44) All-Sky Solar Radiation	RadAvg		1.31	2.25	3.49	5.10	6.19	7.13	7.71	6.52	4.82	2.97	1.60	1.05		
	RadStd		0.14	0.27	0.30	0.35	0.50	0.38	0.27	0.36	0.31	0.22	0.13	0.12		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page